Untitled

Load dataset

```
library(gtsummary)
library(tidyr)
library(data.table)
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:data.table':

    between, first, last

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(data.table)

# Define the base URL for the MSK-CHORD GitHub repository
base_url <- "https://media.githubusercontent.com/media/cBioPortal/datahub/refs/heads/master/p

# Load the datasets directly from GitHub:
clin_patient <- fread(paste0(base_url, "data_clinical_patient.txt"), skip = 4)
# PATIENT_ID, CANCER_TYPE

clin_sample  <- fread(paste0(base_url, "data_clinical_sample.txt"), skip = 4)
# OS_MONTHS, OS_STATUS, CURRENT_AGE_DEID, GENDER, STAGE_HIGHEST_RECORDED, SMOKING_PREDICTIONS

muta <- fread(paste0(base_url, "data_mutations.txt"))
# TP53, KRAS, EGFR (canonical genes)
# ALK, BRAF, STK11, KEAP1, PIK3CA, MET, ERBB2, ROS1 (driver panel)

muta_with_patient <- merge(
    muta,
    clin_sample[, .(SAMPLE_ID, PATIENT_ID, CANCER_TYPE)],
    by.x = "Tumor_Sample_Barcode",
    by.y = "SAMPLE_ID",
    all.x = TRUE
  )

treat <- fread(paste0(base_url, "data_timeline_treatment.txt"))

cancer_pred <- fread(paste0(base_url, "data_timeline_cancer_presence.txt"))
cancer_diag <- fread(paste0(base_url, "data_timeline_diagnosis.txt"))
```

NSCLC

Data cleaning

1. keep only the dataset and rows that related to nsclc

```r
# Select patient ID with non-small cell lung cancer
pID_nsclc <- unique(clin_sample$PATIENT_ID[clin_sample$CANCER_TYPE == "Non-Small Cell Lung Ca

clin_patient_nsclc <- clin_patient %>% filter(PATIENT_ID %in% pID_nsclc)
# GENDER + CURRENT_AGE_DEID + SMOKING_PREDICTIONS_3_CLASSES
# OS_MONTHS + OS_STATUS + PRIOR_MED_TO_MSK
```

```r
clin_sample_nsclc <- clin_sample %>% filter(CANCER_TYPE == "Non-Small Cell Lung Cancer")

cancer_diag_nsclc <- cancer_diag %>% filter(PATIENT_ID %in% pID_nsclc)
# START_DATE + EVENT_TYPE + STAGE_CDM_DERIVED

# Check duplication:
test <- cancer_diag_nsclc[duplicated(cancer_diag_nsclc$PATIENT_ID) | duplicated(cancer_diag_

# 1. have multiple records
# 2. In DX_DESCRIPTION there is no "LUNG";
# 3. If a patient has multiple records after the first two filters, keep only the first one 

cancer_diag_nsclc <- cancer_diag_nsclc %>%
  group_by(PATIENT_ID) %>%
  # 1 & 2. Eliminate records if the patient has > 1 record AND "LUNG" isn't in the descripti
  # We use filter() to keep the rows that DON'T meet your "unwanted" criteria
  filter(!(n() > 1 & !grepl("LUNG", DX_DESCRIPTION, ignore.case = TRUE))) %>%
  # 3. If a patient still has multiple records (e.g., two different "LUNG" entries),
  # keep only the one with the smallest (earliest) START_DATE
  arrange(START_DATE, .by_group = TRUE) %>%
  slice(1) %>%
  ungroup()

muta_with_patient_nsclc <-  muta_with_patient %>% filter(PATIENT_ID %in% pID_nsclc)
# Hugo_Symbol + CANCER_TYPE
```

2. calculate the time variable and event variable needed for the cox model (only clinical features)

```r
# make event variable as 0 or 1 (censored/alive or dead)
nsclc_combined <- clin_patient_nsclc %>%
  inner_join(cancer_diag_nsclc, by = "PATIENT_ID") %>%
  mutate(event = ifelse(grepl("1:DECEASED", OS_STATUS), 1, 0))

nsclc_combined <- nsclc_combined %>%
  mutate(
    age_at_diagnosis = CURRENT_AGE_DEID - (abs(START_DATE) / 365.25),

    entry_time = ifelse(START_DATE < 0, abs(START_DATE), 0) / 30.44,
    exit_time = entry_time + OS_MONTHS
  ) %>%
```

```r
  filter(exit_time > entry_time, !is.na(entry_time), !is.na(exit_time))

# Check missingness after the cleaning
nsclc_combined %>%
  select(everything()) %>%  # replace to your needs
  summarize(across(everything(), ~ sum(is.na(.))))
```

```
  PATIENT_ID GENDER RACE ETHNICITY CURRENT_AGE_DEID STAGE_HIGHEST_RECORDED
1          0      0    0         0                0                      0
  NUM_ICDO_DX ADRENAL_GLANDS BONE CNS_BRAIN INTRA_ABDOMINAL LIVER LUNG
1           0              0    0         0               0     0    0
  LYMPH_NODES OTHER PLEURA REPRODUCTIVE_ORGANS SMOKING_PREDICTIONS_3_CLASSES
1           0     0      0                   0                             0
  GLEASON_FIRST_REPORTED GLEASON_HIGHEST_REPORTED HISTORY_OF_PDL1
1                   7628                     7628               0
  PRIOR_MED_TO_MSK OS_MONTHS OS_STATUS HR HER2 START_DATE STOP_DATE EVENT_TYPE
1                0         0         0  0    0          0      7802          0
  SUBTYPE SOURCE DX_DESCRIPTION AJCC CLINICAL_GROUP PATH_GROUP
1       0      0              0    0              0          0
  STAGE_CDM_DERIVED SUMMARY event age_at_diagnosis entry_time exit_time
1                 0       0     0                0          0         0
```

3. Categorical variable adjustment

```r
# Select the clinical information will be used
nsclc_combined_clin <- nsclc_combined %>% select(
  PATIENT_ID, GENDER, age_at_diagnosis, STAGE_CDM_DERIVED, SMOKING_PREDICTIONS_3_CLASSES,
  PRIOR_MED_TO_MSK, event, entry_time, exit_time
)

# Adjust categorical variables
nsclc_combined_clin <- nsclc_combined_clin %>%
  mutate(
    smoke_f = factor(SMOKING_PREDICTIONS_3_CLASSES),
    stage_f = factor(STAGE_CDM_DERIVED),
    prior_med_to_msk_f = factor(PRIOR_MED_TO_MSK)
  )
```

4. Add core genomic information:

```r
genomic_features <- muta_with_patient_nsclc %>%
  mutate(mutation = 1) %>% # create a binary indicator (1 = mutated)
  # keep only unique patient-gene pairs
  distinct(PATIENT_ID, Hugo_Symbol, .keep_all = T) %>%
  pivot_wider(
    id_cols = PATIENT_ID,
    names_from = Hugo_Symbol,
    values_from = mutation,
    values_fill = 0 # fills missing genes with 0 if the patient is in this list
  )

# Merge the clinical data with genomic dataset
nsclc_final_nsclc <- nsclc_combined_clin %>%
  left_join(genomic_features, by = "PATIENT_ID") %>%
  mutate(across(c(KRAS, EGFR, TP53), ~replace_na(., 0)))
```

Prelimary presentation:

1. summary table for clinical features for nsclc

```r
# Create a summary table for clinical features
table_clinical <- nsclc_combined_clin %>%
  select(GENDER, age_at_diagnosis, STAGE_CDM_DERIVED, SMOKING_PREDICTIONS_3_CLASSES, PRIOR_ME
  tbl_summary(
    by = GENDER, # Optional: Split the table by Gender to show balance
    label = list(
      age_at_diagnosis ~ "Age at Diagnosis",
      STAGE_CDM_DERIVED ~ "Stage at Diagnosis",
      SMOKING_PREDICTIONS_3_CLASSES ~ "Smoking Status",
      PRIOR_MED_TO_MSK ~ "Prior Medication Status"
    ),
    statistic = list(all_continuous() ~ "{median} ({p25}, {p75})"),
    missing = "ifany"
  ) %>%
  add_overall() %>%
  bold_labels() %>%
  modify_header(label = "**Patient Demographics**")

# To view it in RStudio
table_clinical
```

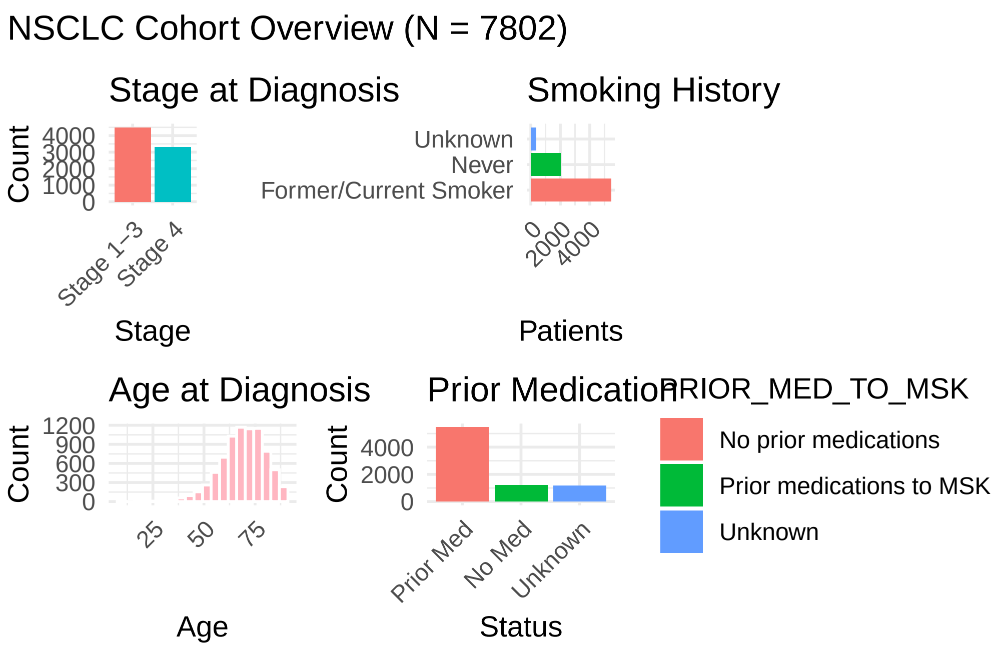
Patient Demographics	**Overall** N = 7,802[1]	**Female** N = 4,565[1]	**Male** N = 3,237[1]
Age at Diagnosis	70 (63, 77)	70 (62, 77)	71 (63, 77)
Stage at Diagnosis			
Stage 1-3	4,497 (58%)	2,655 (58%)	1,842 (57%)
Stage 4	3,305 (42%)	1,910 (42%)	1,395 (43%)
Smoking Status			
Former/Current Smoker	5,434 (70%)	2,959 (65%)	2,475 (76%)
Never	2,025 (26%)	1,416 (31%)	609 (19%)
Unknown	343 (4.4%)	190 (4.2%)	153 (4.7%)
Prior Medication Status			
No prior medications	5,462 (70%)	3,158 (69%)	2,304 (71%)
Prior medications to MSK	1,190 (15%)	685 (15%)	505 (16%)
Unknown	1,150 (15%)	722 (16%)	428 (13%)

[1]Median (Q1, Q3); n (%)

2. clinical features graph

```r
library(patchwork)

# Plot A: Age Distribution
p1 <- ggplot(nsclc_combined_clin, aes(x = age_at_diagnosis)) +
  geom_histogram(fill = "lightpink", color = "white", bins = 20) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1)) +
  labs(title = "Age at Diagnosis", x = "Age", y = "Count")

# Plot B: Smoking Status (Bar Chart)
p2 <- ggplot(nsclc_combined_clin, aes(x = SMOKING_PREDICTIONS_3_CLASSES, fill = SMOKING_PREDI
  geom_bar() +
  coord_flip() + # Makes it easier to read labels
  theme_minimal() +
  theme(legend.position = "none") +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1)) +
  labs(title = "Smoking History", x = "", y = "Patients")

# Plot C: Cancer Stage
p3 <- ggplot(nsclc_combined_clin, aes(x = STAGE_CDM_DERIVED, fill = STAGE_CDM_DERIVED)) +
  geom_bar() +
  theme_minimal() +
  theme(legend.position = "none") +
```

```r
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1)) +
  labs(title = "Stage at Diagnosis", x = "Stage", y = "Count")


p4 <- ggplot(nsclc_combined_clin, aes(x = PRIOR_MED_TO_MSK, fill = PRIOR_MED_TO_MSK)) +
  geom_bar() +
  scale_x_discrete(labels = c("Prior Med", "No Med", "Unknown"))+
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1)) +
  labs(title = "Prior Medication", x = "Status", y = "Count")

# Combine them into a layout
dashboard <- (p3 | p2) / (p1 | p4)
dashboard + plot_annotation(title = "NSCLC Cohort Overview (N = 7802)")
```




```r
# Write data to folder
#library(readr)
#write.csv(nsclc_final_nsclc, "nsclc.csv")
```